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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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ABSTRACT

This proposal outlines the development of **Taranga**, an AI-powered web-based system designed to screen for multiple learning disabilities and suggest personalized, appropriate interventions for Nepali students. Learning difficulties such as dyslexia, dysgraphia, dyscalculia, auditory processing disorder, and nonverbal learning disorder often go undetected in Nepal due to stigma, lack of awareness, and insufficient access to clinical professionals. Taranga aims to address this gap through an adaptive questionnaire system, multi-label machine learning models, and an NLP-based observation analyzer.

The system will recommend targeted strategies and include interactive activities that track student progress over time. To be built using FastAPI, ReactJS, and SQLAlchemy, the project follows a Hybrid Iterative-Waterfall methodology enhanced with prototyping, allowing structured documentation with flexible feature refinement. The system is explicitly non-diagnostic and designed for educational awareness and support. This proposal details the project's objectives, technical design, methodology, risk management, resource planning, and expected outcomes.

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1. Introduction

Millions of children around the world suffer from learning disabilities and most cases go undiagnosed in developing countries. In Nepal, awareness is extremely low, and students with learning difficulties are often neglected. Due to the absence of early screening mechanisms, it is tough for teachers and parents to identify the actual cause behind a child's learning problem. Lack of early detection can result in long-term effects such as poor academic performance, lower self-esteem, dropout rates, and emotional problems.

While educational technology is on the rise across the world, Nepal lags in the required digital tools to help schools identify early signs of difficulties in learning and provide intervention strategies correspondingly. **Taranga** seeks to fill this gap with the provision of an AI-driven early screening and personalized recommendation for intervention in children with symptoms of learning difficulties.

1.1 Problem Scenario

Learning disabilities affect a significant portion of learners worldwide, with developmental learning disorders impacting 5 to 15% of school-aged children (American Psychiatric Association, n.d.). Dyslexia, one of the most common learning difficulties, impacts a substantial portion of the population, with studies suggesting that 15 to 20% of individuals may show symptoms of dyslexia (International Dyslexia Association , n.d.). In developing countries, however, these conditions often remain undetected due to stigma and limited awareness.

In Nepal, similar challenges are evident. A study conducted in Madi Municipality, Chitwan, found that the majority of primary school teachers had limited knowledge and no formal training related to learning disabilities. Only 9.5% demonstrated adequate knowledge, while most had moderate or inadequate understanding (Thapa, et al., 2024). As a result, students struggling with reading, writing, or attention are frequently mislabelled as "slow" or "undisciplined" rather than being screened for LDs. Human Rights Watch (2018) further reports a severe shortage of trained teachers, and UNICEF Nepal (2021) notes that the

country lacks standardized screening tools and structured early-intervention processes. These findings reveal a clear gap in early detection and structured support.

1.2 Project as a Solution

Taranga proposes an AI-powered screening and recommendation system designed specifically for the Nepali educational context. The system provides:

- Adaptive questionnaires that adjust based on responses, enabling more accurate screening.
- Multi-label classification to identify symptoms across several learning disabilities simultaneously.
- Teacher Observation Analyzer using NLP to extract patterns and indicators of LD from teacher notes.
- Personalized intervention recommendations suggesting classroom strategies, home-based remedies, and learning supports.
- AI explainability feature that help teachers and parents understand why the system flagged certain indicators, promoting transparency and trust in AI-driven decisions.

2 Aims and Objectives

Aim:

To develop an AI-powered multidimensional screening and intervention recommendation system that identifies learning disabilities early and supports Nepali students through personalized, data-driven strategies.

2.1 Business / Project Objectives

- To provide an early-screening platform that identifies symptoms of major learning disabilities among Nepali students.
- To support inclusive education practices by offering clear, structured reports that help teachers and parents understand a student's learning patterns and challenges.

2.2 Technical / System Objectives

- To design and implement an adaptive questionnaire module that adjusts questions dynamically based on responses to improve screening accuracy for the five LD categories.
- To develop a multi-label machine learning model capable of predicting the likelihood of one or more learning disabilities based on questionnaire responses and behavioral patterns.
- To create an NLP-based teacher observation analyzer that extracts early-warning indicators of learning disabilities from qualitative teacher notes.
- To integrate a personalized intervention recommendation engine that generates remedies, accommodations, and support strategies tailored to each LD category.
- To develop interactive intervention activities for each LD type and implement a mechanism to track student performance and improvement over time.
- To implement AI explainability mechanisms for transparency and trust in AI-driven predictions.
- To build a secure, user-friendly web platform with role-based access control for teachers, parents, and administrators using FastAPI, React, and SQLAlchemy.

- To develop a structured database schema for efficient storage, retrieval, and analysis of student data, screening results, and intervention progress.

2.3 Personal Development Objectives

- To gain hands-on experience in AI/ML technologies, including multi-label classification, NLP, explainability techniques, and model evaluation.
- To improve full-stack development skills through the use of FastAPI, ReactJS, and SQLAlchemy for building a production-ready system.
- To enhance software engineering competencies, including project documentation, methodology selection, testing, deployment, and version control.

3 Expected Outcomes and Deliverables

The project is expected to produce a fully functional, web-based multidimensional learning disability screening and intervention support system designed for use in Nepali schools. The primary outcome will be a platform capable of identifying early indicators of learning disabilities and recommending suitable interventions to support students, teachers, and parents.

The expected deliverables include:

3.1 Functional Web Application

A complete web system developed using FastAPI, ReactJS, and SQLAlchemy that provides:

- Adaptive screening questionnaires
- Multi-label LD prediction results
- NLP-based teacher observation analysis
- Personalized intervention recommendations
- Interactive practice activities for each LD type
- Student progress tracking
- AI explainability feature
- Role-based access for teachers, parents, and administrators

3.2 Machine Learning and NLP Models

Deliverables will include trained models for:

- Multi-label LD classification
- NLP processing of teacher notes
- Recommendation ranking or retrieval logic
- Explainable AI outputs

3.3 Technical Documentation

Comprehensive documentation covering:

- System architecture
- Database schema
- APIs and endpoints
- Model training workflow
- Explainability methods
- Deployment process

3.4 User Documentation

Clear user guides and instructions for teachers, parents, and school administrators to navigate the system.

3.5 Project Report

A complete academic report documenting the research background, methodology, implementation, evaluation, limitations, and future improvements.

4 Project risks, threats and contingency plans

Risk ID	Risk Description	Impact (1–5)	Likelihood (1–5)	Risk Level	Mitigation Strategy	Contingency Plan
1	AI Model Misclassification: Incorrect learning difficulty predictions due to limited/simulated dataset.	4	4	16	Validate model iteratively, combine rule-based and ML approach, use explainability to show reasoning.	Display confidence scores. allow retakes, emphasise non-diagnostic purpose in UI.
2	NLP Misinterpretation of Teacher Observations: Free-text input may lead to inaccurate symptom extraction.	3	4	12	Use keyword-based fallback. limit NLP to non-clinical insights only, test with varied text samples.	Allow manual edits of extracted symptoms, show disclaimer for NLP accuracy.
3	Integration Failures Between Modules: Issues between frontend, backend, and AI model endpoints.	4	3	12	Adopt modular API design, perform integration tests, document API contracts.	Use mocked API responses until issues are resolved.
4	Database Corruption or Loss of Screening Logs: Loss of student progress data.	4	3	12	Perform automated backups, enforce DB constraints and transactions.	Restore latest backup, provide read-only mode while recovering.

5	Performance Issues: Slow prediction time for ML/NLP models or slow API responses.	3	3	9	Optimise queries, preload ML model on server startup, implement caching.	Switch to rule-based fallback model temporarily.
6	User Interface Confusion: Teachers or parents may misunderstand screening flow or analytics.	2	3	6	Conduct usability tests, simplify layout, add clear labels and instructions.	Add guided tutorial mode and contextual tooltips.
7	Ethical Misuse or Misinterpretation: Users treat Taranga as a medical diagnosis tool.	5	3	10	Display disclaimers, use non-clinical terminology, educate users within onboarding.	Include referral information to professionals, require confirmation before final results appear.

Table 1 Risks, Threats and Contingency plans

Risk Level Indicators:

Very Low Risk (1, 1-5)

Low Risk (2, 6-10)

Medium Risk (3, 11-15)

High Risk (4, 16-20)

Very High Risk (5, 21-25)

5 Methodology

This project follows a Hybrid Iterative–Waterfall methodology enhanced with Prototyping, providing a balance between structured SDLC phases and the flexibility required for AI model development and user interface refinement. As the project is undertaken individually and within a fixed timeline, a full Agile approach is unsuitable. However, the complexity of machine learning, NLP, and adaptive UI components requires iterative cycles for experimentation and improvement. The hybrid model ensures clear planning while enabling controlled iteration.

5.1 Overview of the SDLC Approach

5.1.1 Requirements Analysis

This phase involves identifying and documenting both functional and non-functional requirements for Taranga. It includes researching learning disabilities (based on DSM-5 and psychological literature), evaluating existing screening tools, and identifying gaps in the Nepali education context. Role-specific requirements (for teachers, parents, and admins), accessibility, data privacy, ML interpretability, and intervention mechanisms are all covered in this phase.

5.1.2 System & Model Design (with Prototyping)

Early-stage prototyping will be used to validate critical design assumptions. Low-fidelity wireframes for the dashboard, adaptive questionnaire flow, and progress tracker will be created and iteratively improved. At the same time, the technical design includes:

- Database schema (ER diagrams)
- API structure and request flows
- ML model structure
- NLP data extraction logic
- User interface layouts (based on user roles)

5.1.3 Implementation (Iterative Cycles)

The core system will be developed in five short iterative cycles, each concluding with internal testing and feedback review.

- **Iteration 1**: Adaptive questionnaire engine (dynamic flow logic, role-specific question filters, and response storage)
- **Iteration 2**: Multi-label machine learning model (initially rule-based, then trained on mock data for LD prediction)
- **Iteration 3**: NLP-based teacher observation analyzer (keyword extraction, basic entity recognition, free-text analysis)
- **Iteration 4**: Intervention recommendation engine and activity tracker (LD-wise activities, scoring metrics, and activity log storage)
- **Iteration 5**: Role-based access control, dashboard visualization, and full progress tracker integration

5.1.4 Integration & Testing (Final System Validation)

Once all modules are individually developed, they will be fully integrated and tested in a local environment. Testing will include:

- Unit testing of Python and React components
- Validation of adaptive logic and question flows
- API testing with FastAPI
- Evaluation of ML/NLP modules using accuracy, recall, and explainability checks
- Usability validation through simulated user interactions (teacher/parent use cases)
- Cross-checking data consistency and logic integrity

5.1.5 Evaluation & Documentation

The final system will be evaluated based on:

- AI model performance (precision, recall, multi-label accuracy)
- NLP analyzer reliability

- User engagement flow (questionnaire to intervention)
- Progress tracking reliability

Comprehensive technical documentation, user instructions, and annotated screenshots will be produced. Limitations, ethical considerations, and future improvement paths will also be outlined in the final report.

6 Resource Requirements

Category	Resource	Purpose / Description
Hardware	MacBook Pro (14-inch, 2023, Apple M2 Pro, 16 GB RAM)	Main development machine to be used throughout the project lifecycle.
Software & Tools	Python	Core language to be used for backend logic, ML pipeline, and NLP modules.
	FastAPI	High-performance backend framework used for API development and serving AI predictions.
	ReactJS	Frontend library for building the questionnaire, dashboards, and intervention interface.
	SQLAlchemy	ORM for Python to interact with PostgreSQL database securely and efficiently.
	PostgreSQL	Relational database to store everything necessary.
	VS Code	IDE for coding, debugging, and version control integration.
	Git & GitHub	To be used for version control, project backup, issue tracking and documentation logs.
	Jupyter Notebook/ Google Colab	To be used for prototyping, training and evaluating ML models and NLP workflows interactively.
	Microsoft Word / Google Docs	To be used for writing all the documentation during the project
	draw.io / diagrams.net	For designing system architecture, UML diagrams, and ERDs.

	Figma / Canva	To be used for UI mockups and wireframes.
	GanttPRO and Trello	Project timeline planning, Gantt chart creation, and WBS visualization.
AI/ML Libraries	scikit-learn, pandas, NumPy	To be used for multi-label classification, dataset processing, and exploratory analysis.
	spaCy / NLTK / HuggingFace Transformers	NLP libraries for implementing the teacher observation analyzer, keyword extraction, and entity recognition.
	SHAP / LIME	Explainable AI tools to help make ML model predictions interpretable, especially for LD prediction rationale.
Datasets & Research	Mock/Synthetic LD datasets (custom-generated)	Used to train the multi-label classifier in the absence of real-world diagnostic data. Simulated responses based on DSM-5 symptoms.
	DSM-5, Academic Journals, Research Articles	Used to derive questionnaire logic, LD categorization, intervention strategies, and screening behaviour models.

Table 2 Resource Requirement Table

7 Work Breakdown Structure

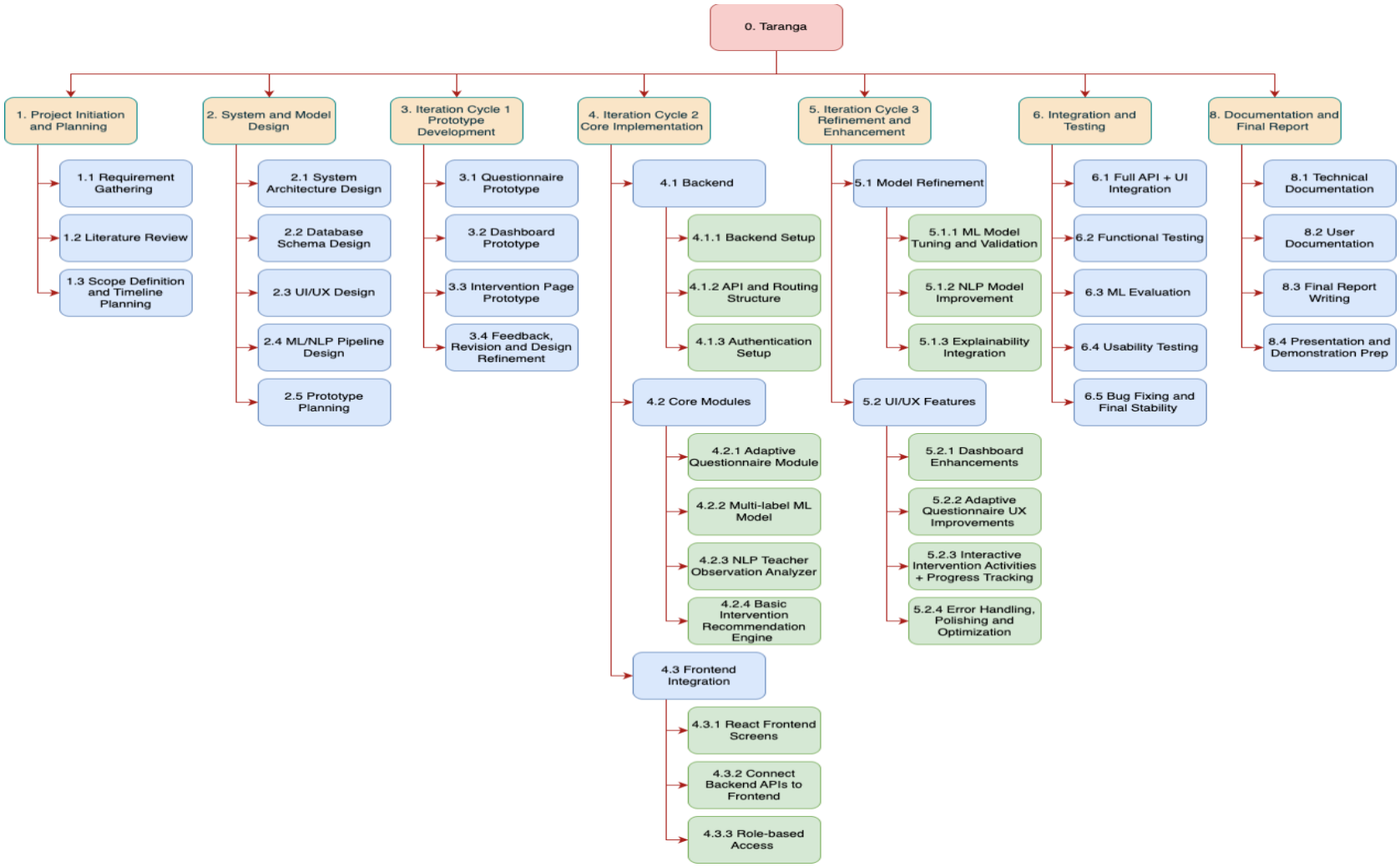

Figure 1 WBS

8 Milestones

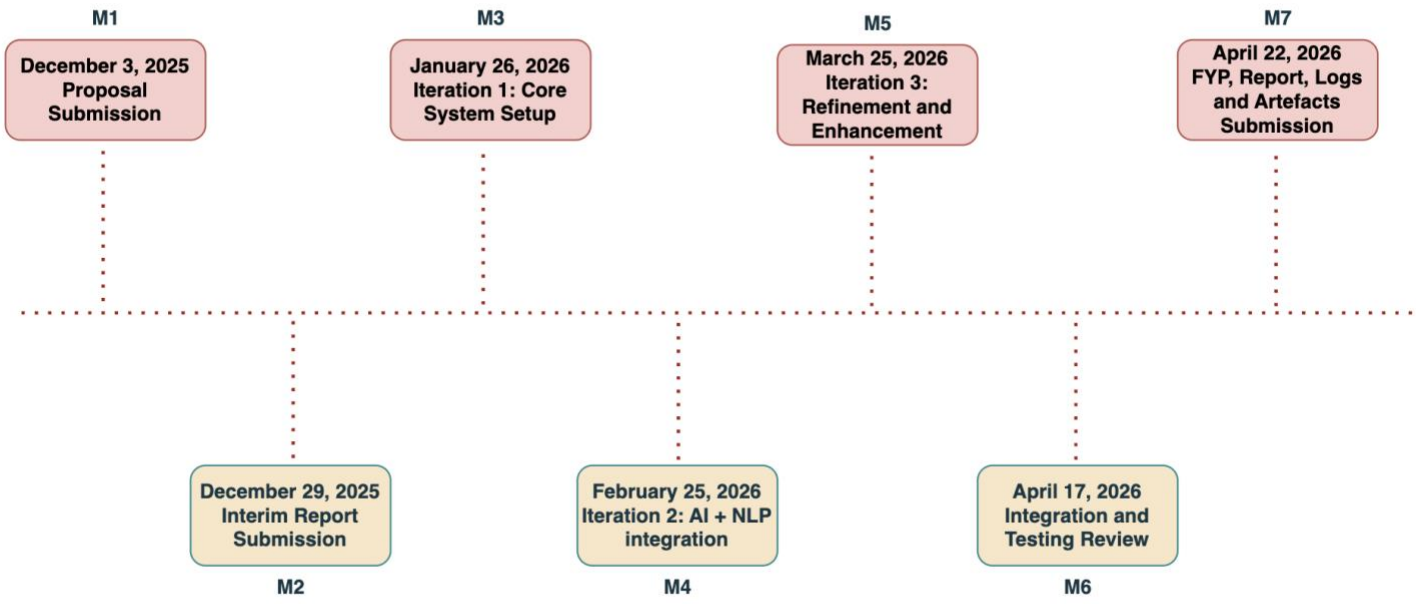

Figure 2 Milestones

9 Project Gantt Chart

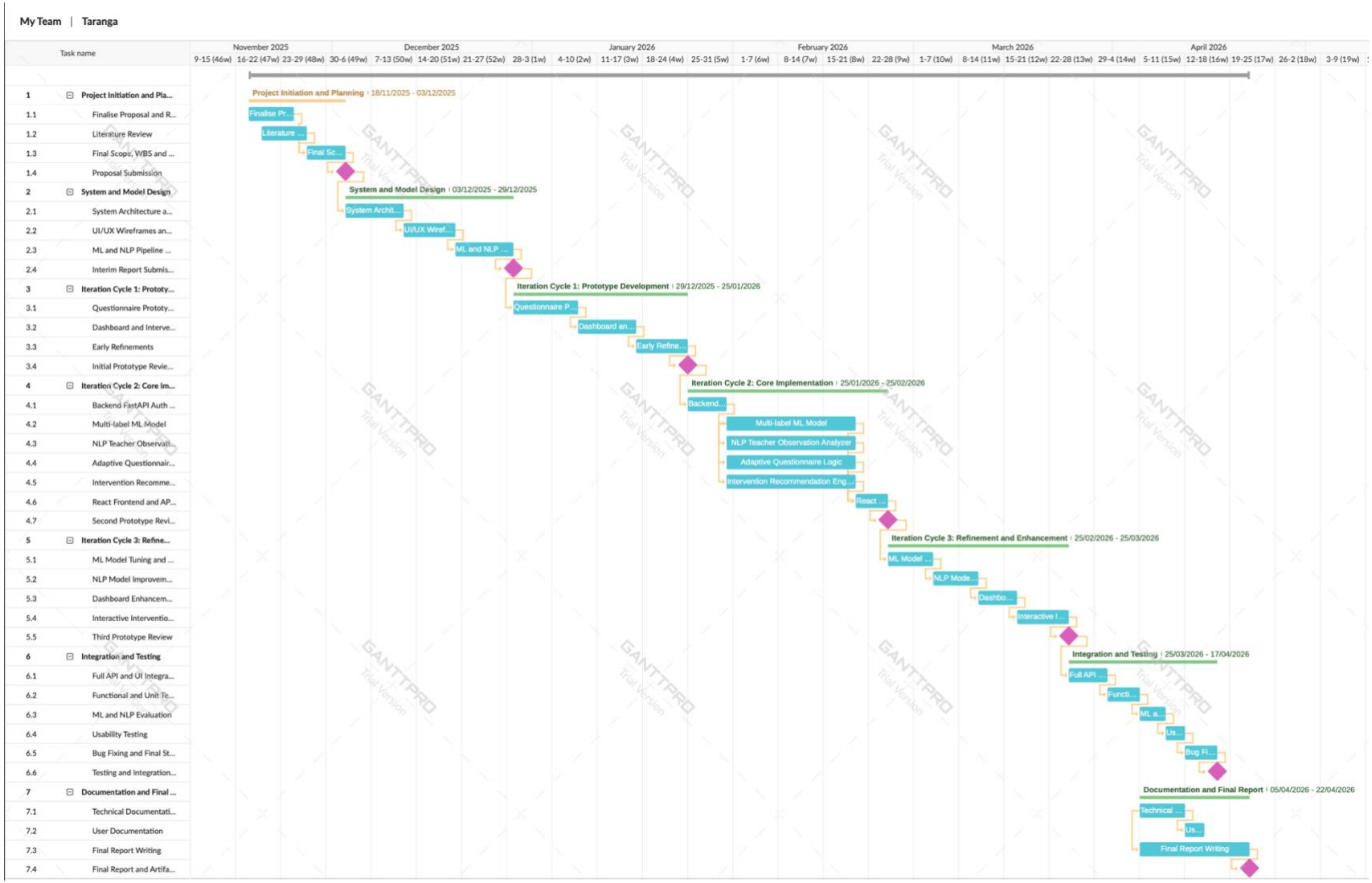

Figure 3 Gantt Chart

10 Conclusion

Taranga is envisioned as a socially responsible, AI-assisted screening and support platform for early identification of learning disabilities among school-aged children in Nepal. With its adaptive, data-driven approach and integration of local context, the system strives to empower teachers and parents with awareness and actionable strategies rather than clinical diagnoses.

By following a hybrid development methodology and integrating interactive activities with measurable progress tracking, the project aligns both technically and ethically with its purpose. Through careful implementation and testing, Taranga will deliver a meaningful, accessible tool that contributes toward inclusive education and early support systems in under-resourced settings.

This proposal establishes the foundation for successful implementation of the project by detailing the goals, design plans, development strategy, and risk preparedness for the months ahead.

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